



Industries that will grow with EV expansion

- **Component manufacturing.** Especially in India, driven by competitive pricing and faster lead times
- **Engineering and validation services.** Including outsourced simulation and data annotation
- **Charging infrastructure.** Hardware manufacturing, installation networks, and home power systems
- **Fleet connectivity and telematics.** Predictive maintenance, cybersecurity, and over-the-air updates
- **Data and analytics.** Performance monitoring, maintenance optimisation, and customer experience insights

benchmarks that Indian models are yet to match.

There is also room to advance customer experience and vehicle design, particularly in features such as compact headlamps using adaptive matrix LED or micro-lens technologies integrated with electronics, sensors, and advanced optics.

How programme management is powering large-scale EV launches

Strong programme management has emerged as a key differentiator in large-scale EV launches. Traditional automakers such as Ford, General Motors, and Stellantis have historically worked on four- to five-year development cycles, while newer companies like Tesla and Lucid have reduced timelines to two to three years, making structured programme management a critical enabler of competitive market entry.

At its core, effective programme management requires disciplined integration and governance across hardware, software, validation, testing, homologation, and regulatory milestones. It also requires clearly defined programme gates aligned with executive leadership, structured risk management systems with formal escalation paths, and contingency frameworks designed to protect schedule, cost, and quality objectives. Progress is measured not only by gate adherence and schedule performance but also by confirmation that engineering validation, certification, and customer requirements

remain aligned at launch.

Real-world examples show how this discipline supports scale. Rivian's electric delivery van, launched in 2023, was designed as a flexible platform adapted for logistics, emergency services, and other fleet applications. Such expansion depends on data-driven decisions using telematics, demand forecasts, cost reviews, and KPI dashboards to improve rollout speed, maintain transparency, and build customer trust.

Programme management practices differ between startups and established OEMs. Large OEMs typically move from T0 to T3 parts across multiple validation stages prior to homologation, whereas startups often compress this validation pathway and may deploy early-stage parts directly for design verification and certification to accelerate time-to-market. This shortens launch timelines but increases risk, making disciplined supply chain development, strategic outsourcing decisions, and rigorous supplier coordination essential to balancing speed with validation integrity and quality assurance.

Artificial intelligence is reshaping programme management. AI tools, including generative AI, help identify risks and interdependencies early and summarise large volumes of data from design, manufacturing, supply chain, and logistics into clear insights. When integrated with project and analytics platforms such as Jira and Tableau, AI-driven frameworks provide real-time

portfolio visibility and milestone deviation tracking for executive leadership teams. Customised AI models can track specific vehicle categories, such as two- or three-wheelers, and proactively flag deviations from programme baselines to improve reporting speed, risk-detection accuracy, and data-driven decision-making.

Emerging practices further strengthen the execution of the EV programme. Platforming core capabilities enables reuse of hardware architectures, software stacks, and operational playbooks across regions, materially reducing cycle time and cost while preserving validation consistency. Ecosystem partnerships with utilities, component suppliers, validation firms, digital service providers, and fleet connectivity partners are increasingly central to scalable execution under compressed schedules. Strong data governance and unified telematics systems enhance demand forecasting and customer insight. Monitoring early signals of incentive shifts, trade policy adjustments, or evolving technical standards enables programme teams to respond early and reduce schedule disruption.

The global EV market is growing at different speeds across regions. China is leading, the US is slowing, and India is accelerating with affordable models and government support. Making EVs work at scale requires more than vehicles alone; it depends on charging networks, battery innovation, and disciplined programme execution. By combining data, AI, partnerships, and platform-based approaches, companies can launch faster without compromising quality, helping EVs reach more users and support the transition to cleaner transportation. **EFY**

This article is based on an exclusive conversation with Jagruti Dhande, Senior Technical Program Manager (TPM), Lucid Motors. It has been transcribed and edited by Nidhi Agarwal and Shubha Mitra from EFY.